

5.

$$(a) \text{Var}(b_1|X) = \frac{\sigma^2}{\sum (X_t - \bar{X})^2}$$

$$X_t = t \Rightarrow \sum (X_t - \bar{X}) = \sum_{t=1}^T (t - \bar{t})$$

$$= \sum_{t=1}^T t^2 - \frac{(\sum t)^2}{T}$$

$$= \frac{T(T+1)(2T+1)}{6} - \frac{\left[\frac{T(T+1)}{2}\right]^2}{T}$$

$$= \frac{T^3 - T}{12}$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{Var}(b_1|X) = \lim_{T \rightarrow \infty} \frac{\sigma^2}{\frac{T^3 - T}{12}} = 0, \text{ consistent}$$

$$(b) X_t = 0.5^t$$

$$\text{Var}(b_1|X) = \frac{\sigma^2}{\sum (X_t - \bar{X})^2} = \frac{\sigma^2}{\sum X_t^2 - N \bar{X}^2}$$

$$= \frac{\sigma^2}{\sum (0.5^t)^2 - T \times \left(\frac{\sum 0.5^t}{T}\right)^2}$$

$$= \frac{\sigma^2}{\frac{0.25(1 - 0.25^T)}{1 - 0.25} - T \times \left(\frac{0.5(1 - 0.5^T)}{1 - 0.5}\right)^2}$$

$$= \frac{\sigma^2}{\frac{(1-0.25^T)}{3} - \frac{(1-0.5^T)}{T}}$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{Var}(b_1 | X) = \frac{\sigma^2}{\frac{1}{3} - 0} = 3\sigma^2 \neq 0, \text{ is not consistent.}$$

(c)

- ① T 持續增加提供更多 x_t , 並確立 x, y 之間的關係
- ② T 持續增加讓 $x_t \rightarrow 0$, 以致不能提供 β_2 的資訊

b.

(a)

$$\begin{cases} H_0: \beta_2 = 0 \end{cases}$$

$$\begin{cases} H_a: \beta_2 \neq 0 \end{cases}$$

$$\text{test statistic: } t^* = \frac{b_2 - 0}{\text{SE}(b_2)} = \frac{3}{\sqrt{4}} = 1.5$$

$$\text{R.R.} = \{ |T| \geq t_{0.05}(60) = 2 \}$$

$$t^* \notin \text{RR} \Rightarrow \text{do not reject } H_0$$

(b)

$$\begin{cases} H_0: \beta_1 + 2\beta_2 = 5 \end{cases}$$

$$\begin{cases} H_a: \beta_1 + 2\beta_2 \neq 5 \end{cases}$$

$$\text{test statistic: } t^* = \frac{b_1 + 2b_2 - 5}{SE(b_1 + 2b_2)} = \frac{(2+2 \times 3) - 5}{\sqrt{3 + 2^2 \times 4 + 2 \times 2 \times 2}} = \frac{3}{\sqrt{11}}$$

$$R.R. = \{ |T| \geq t_{0.05}(60) \}$$

$t^* \notin R.R. \Rightarrow$ do not reject H_0 .

(c)

$$\begin{cases} H_0: \beta_1 - \beta_2 + \beta_3 = 4 \\ H_a: \beta_1 - \beta_2 + \beta_3 \neq 4 \end{cases}$$

$$\begin{aligned} \text{test statistic: } t^* &= \frac{b_1 - b_2 + b_3 - 4}{SE(b_1 - b_2 + b_3)} \\ &= \frac{(2 - 3 + (-1)) - 4}{\sqrt{3 + 4 + 3 + 2 \times (-1) \times (-1) + 2 \times (-1) \times 2 \times (-1) \times 0}} \\ &= -1.5 \end{aligned}$$

$$R.R. = \{ |T| \geq t_{0.05}(60) \}$$

$t^* \notin R.R. \Rightarrow$ do not reject H_0

20.

$$(a) \begin{cases} \bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1 \\ \bar{y}_v = \beta_1 + \beta_2 \bar{x}_v + \bar{e}_v \end{cases}$$

$$\hat{\beta}_2 \text{ . mean} = \frac{\bar{y}_v - \bar{y}_1}{\bar{x}_v - \bar{x}_1} = \beta_2 + \frac{\bar{e}_v - \bar{e}_1}{\bar{x}_v - \bar{x}_1}$$

$$\Rightarrow E(\hat{\beta}_2 \text{ . mean} | X) = \beta_2 + \frac{E(\bar{e}_v | X) - E(\bar{e}_1 | X)}{\bar{x}_v - \bar{x}_1} = \beta_2$$

$$E(\hat{\beta}_2 \text{ . mean}) = E[E(\hat{\beta}_2 \text{ . mean} | X)] = E(\beta_2) = \beta_2$$

$\therefore \hat{\beta}_{2, \text{mean}}$ 是 β_2 的不偏 estimator.

$$b) \text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \text{Var}\left(\frac{\bar{e}_2 - \bar{e}_1}{(\bar{x}_2 - \bar{x}_1)} | X\right) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \text{Var}(\bar{e}_2 - \bar{e}_1 | X)$$

$$\text{其中 } \text{Var}(\bar{e} | X) = \text{Var}\left(\frac{1}{3} \sum_{i=1}^3 e_{ii} | X\right) = \frac{1}{3^2} \sum_{i=1}^3 \text{Var}(e_{ii} | X) = \frac{\sigma^2}{3}$$

$$\Rightarrow \text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \times \left(\frac{\sigma^2}{3} + \frac{\sigma^2}{3}\right) = \frac{\frac{2}{3} \sigma^2}{(\bar{x}_2 - \bar{x}_1)^2}$$

$$\begin{aligned} c) \text{Var}(\hat{\beta}_{2, \text{mean}}) &= E[\text{Var}(\hat{\beta}_{2, \text{mean}} | X)] + \text{Var}[E(\hat{\beta}_{2, \text{mean}} | X)] \\ &= \frac{2}{3} \sigma^2 E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] + 0 \end{aligned}$$

d) 如果要 consistent, $N \rightarrow \infty$, $\text{Var}(\hat{\beta}_{2, \text{mean}}) \rightarrow 0$

$$\Rightarrow \text{Var}(\hat{\beta}_{2, \text{mean}}) = \frac{4}{N} \sigma^2 E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$$

$$\lim_{n \rightarrow \infty} \frac{4}{N} \sigma^2 E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] = 0$$

⑩ [1] "--- Exercise 5.20 (e) Simulation Results ---"

	N	var_b2_x	var_beta2mean_x	E_inv_sx2	E_formula	results_var_beta2mean_x	
1	100	1.3843415		NA	0.1384341	0.1828591	0.91429560
2	500	0.2451538		NA	0.1225769	0.1668873	0.16688735
3	1000	0.1147848		NA	0.1147848	0.1517362	0.07586811
4	5000	0.0234716		NA	0.1173580	0.1569605	0.01569605

$$\text{if) } \begin{cases} \text{Var}(b_2 | X) = \frac{\sigma^2}{N S_x^2}, & \text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \\ E[(S_x^2)^{-1}] \doteq 0.12, & E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \doteq 0.16 \end{cases}$$

$$\therefore \text{Var}(b_2 | X) \approx \frac{0.12 \times 1000}{N}, \quad \text{Var}(\hat{\beta}_{2, \text{mean}} | X) \approx \frac{0.16 \times 1000}{N}$$

$$\Rightarrow \text{Var}(b_2 | X) < \text{Var}(\hat{\beta}_{2, \text{mean}} | X)$$

② 當 N 增加, 兩者分別往 0.12 及 0.16 靠近

③ $\therefore \kappa_i \sim U(0, 10)$

$\therefore$ 當 $N \rightarrow \infty$, $\bar{\kappa}_1 \xrightarrow{P} 2.5$, $\bar{\kappa}_2 \xrightarrow{P} 7.5$

$$\text{Var}(\hat{\beta}_{2, \text{mean}} | X) \xrightarrow{P} \frac{4\sigma^2}{N \times (10.1)^2} \rightarrow 0$$

g) $E[(S_x^2)^{-1}] \xrightarrow{P} 0.12$

$$S_x^2 \xrightarrow{P} \text{Var}(X) = \frac{(10-0)^2}{12}$$

$$(S_x^2)^{-1} \xrightarrow{P} \frac{2}{100} = 0.12$$

$$E\left[\frac{4}{(\bar{\kappa}_2 - \bar{\kappa}_1)^2}\right] \xrightarrow{P} 0.16$$

前半段 $U(0, 5)$

後半段 $U(5, 10)$

$$\bar{\kappa}_1 \xrightarrow{P} 2.5$$

$$\bar{\kappa}_2 \xrightarrow{P} 7.5$$

$$\bar{\kappa}_2 - \bar{\kappa}_1 \xrightarrow{P} 7.5 - 2.5 = 5$$

$$\Rightarrow \frac{4}{(\bar{\kappa}_2 - \bar{\kappa}_1)^2} \xrightarrow{P} \frac{4}{25}$$

h) 若非上升而是隨機, $\bar{\kappa}_1 \xrightarrow{P} 5$, $\bar{\kappa}_2 \xrightarrow{P} 5$

$$\bar{\kappa}_2 - \bar{\kappa}_1 \xrightarrow{P} 0$$

$$\Rightarrow \text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \frac{4\sigma^2}{N(\bar{\kappa}_2 - \bar{\kappa}_1)^2} \text{無法收斂到 } 0$$

故 $\hat{\beta}_{\text{mean}}$ would not be consistent